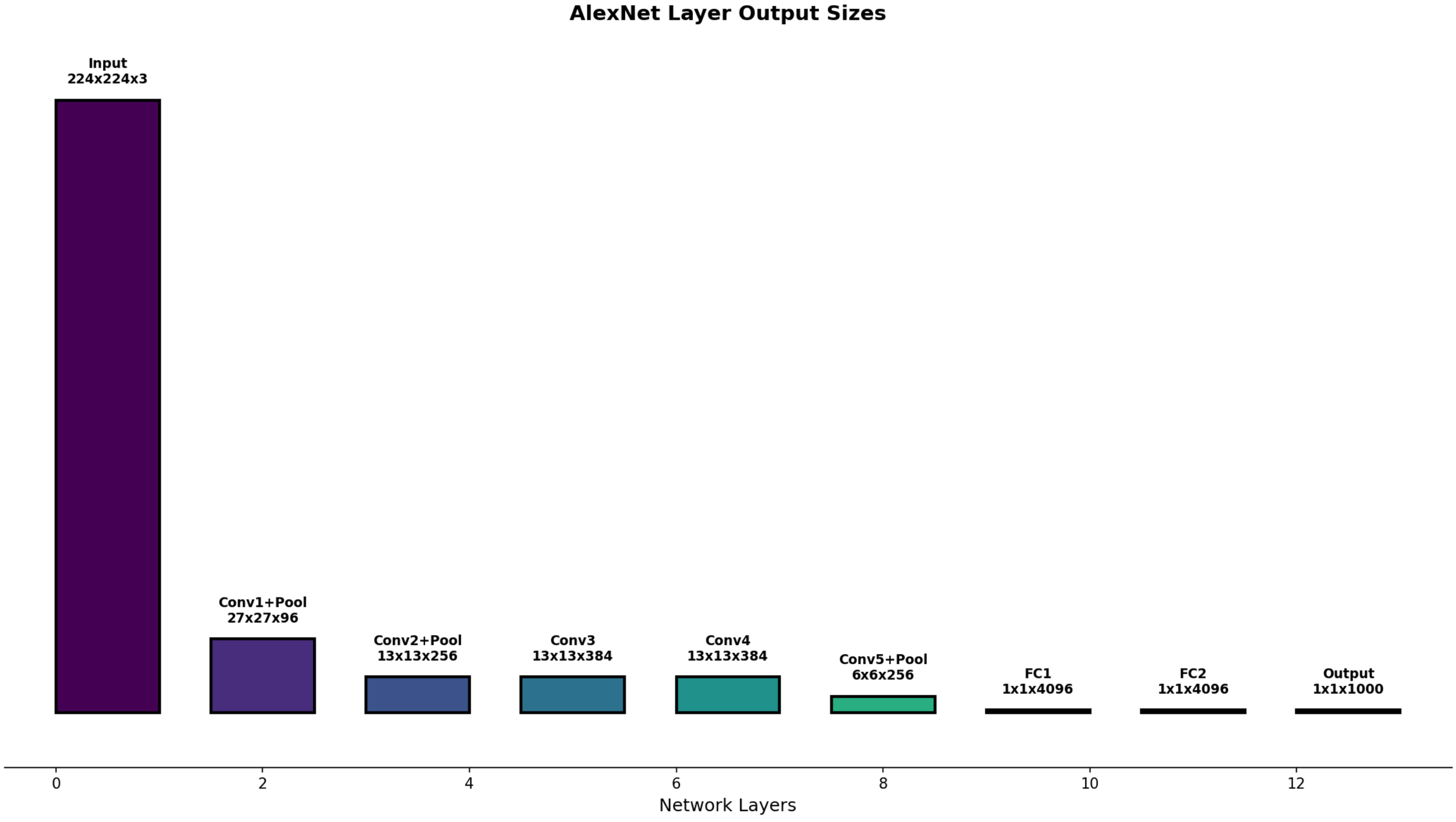
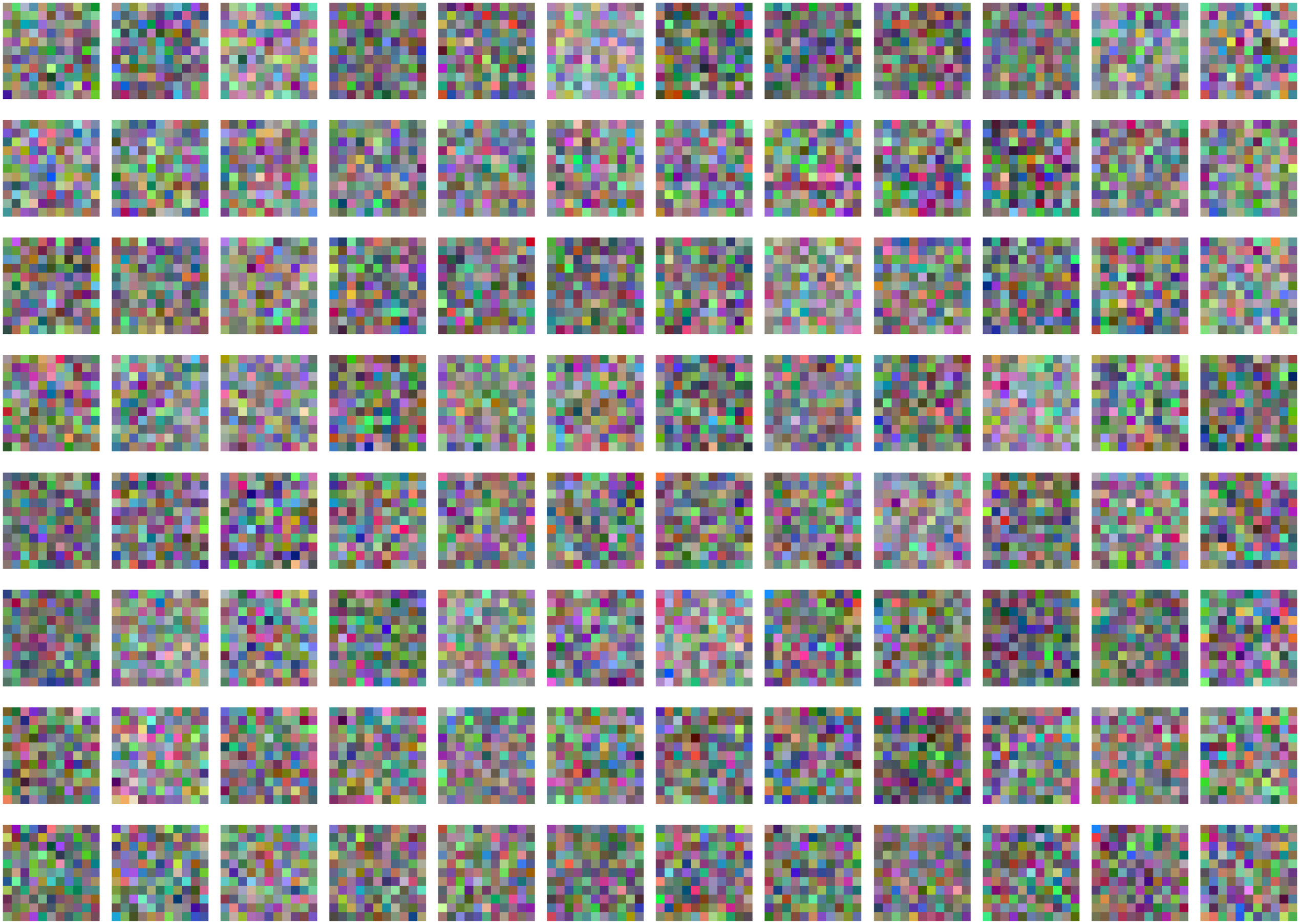
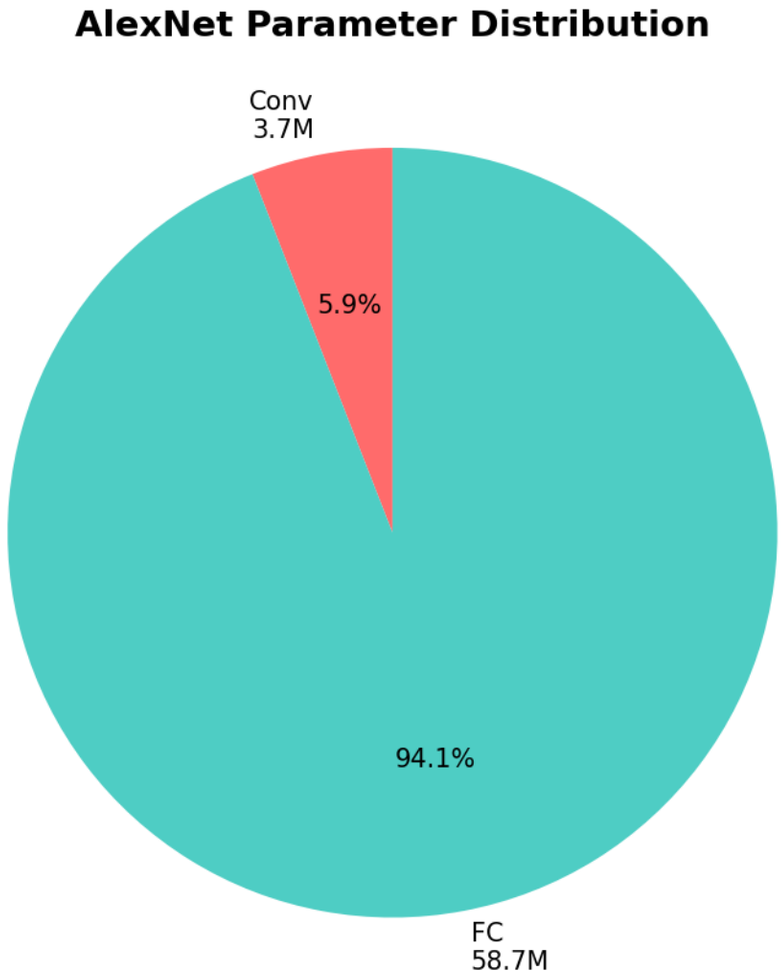
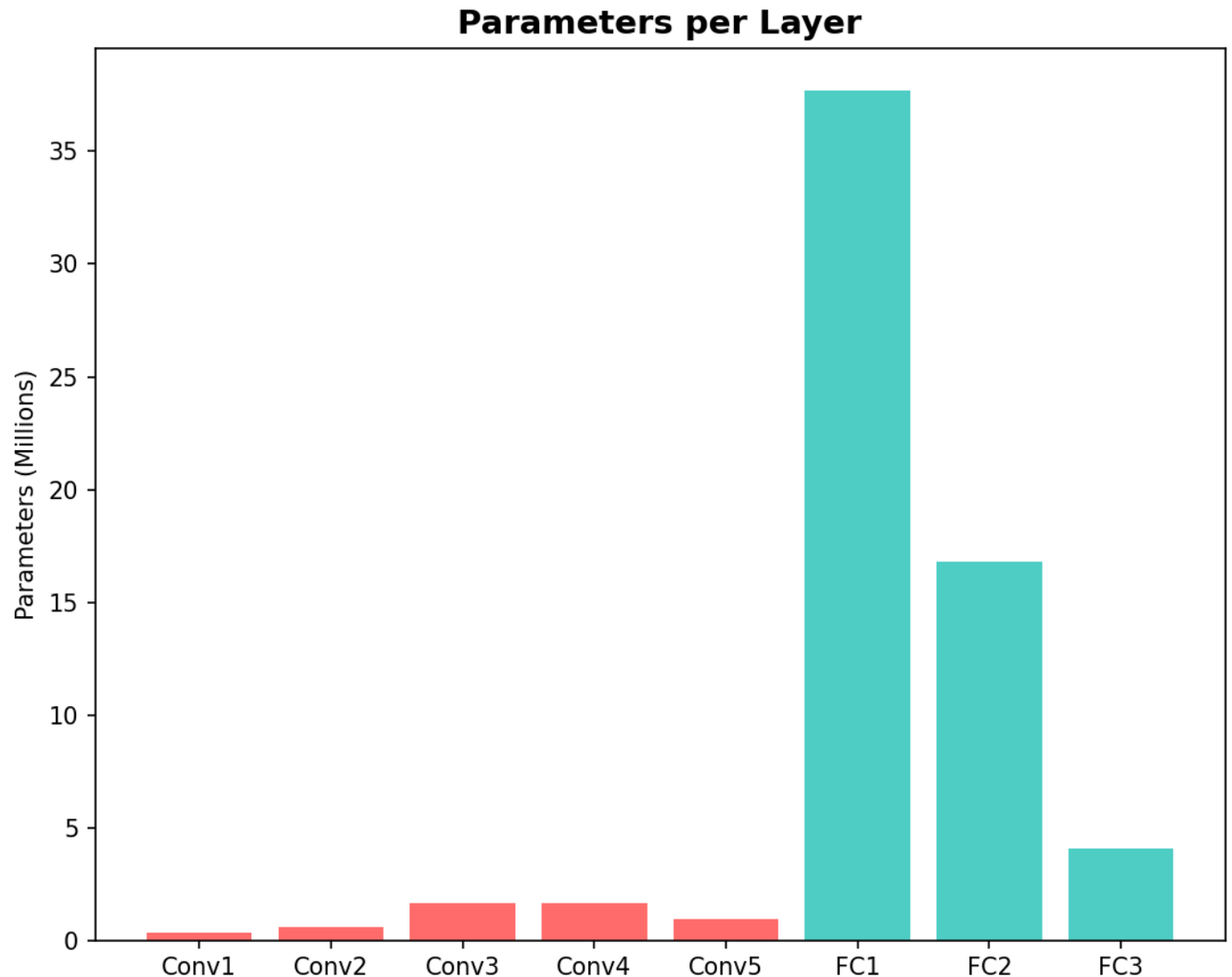

AlexNet Layer Output Sizes
Input
224x224x3
Conv1+Pool
27x27x96
Conv2+Pool
13x13x256
Conv3
13x13x384
Conv4
13x13x384
Conv5+Pool
6x6x256
FC1
1x1x4096
FC2
1x1x4096
Output
1x1x1000
0
2
4
6
8
10
12
Network Layers

AlexNet Conv1 Kernels (96 filters, 11x11 each)

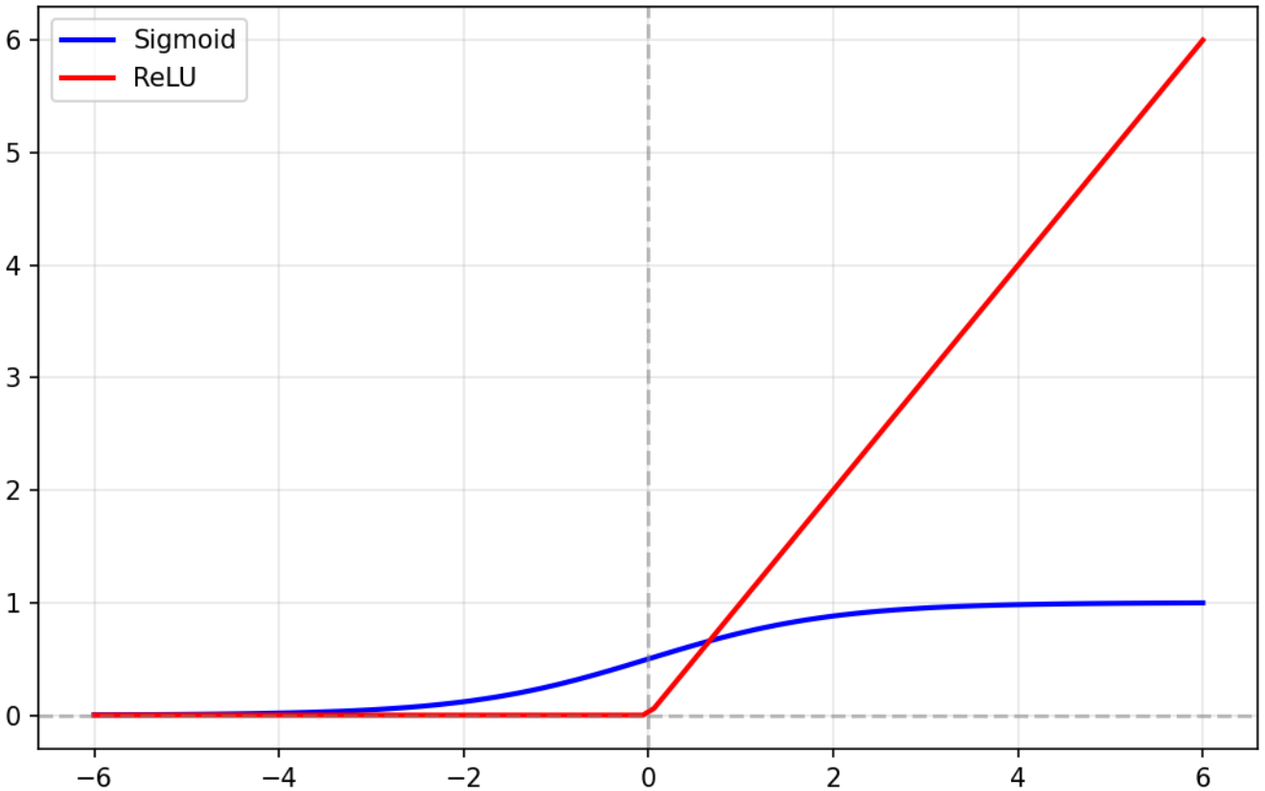



AlexNet Parameter Distribution
Conv
3.7M
5.9%
94.1%
FC
58.7M

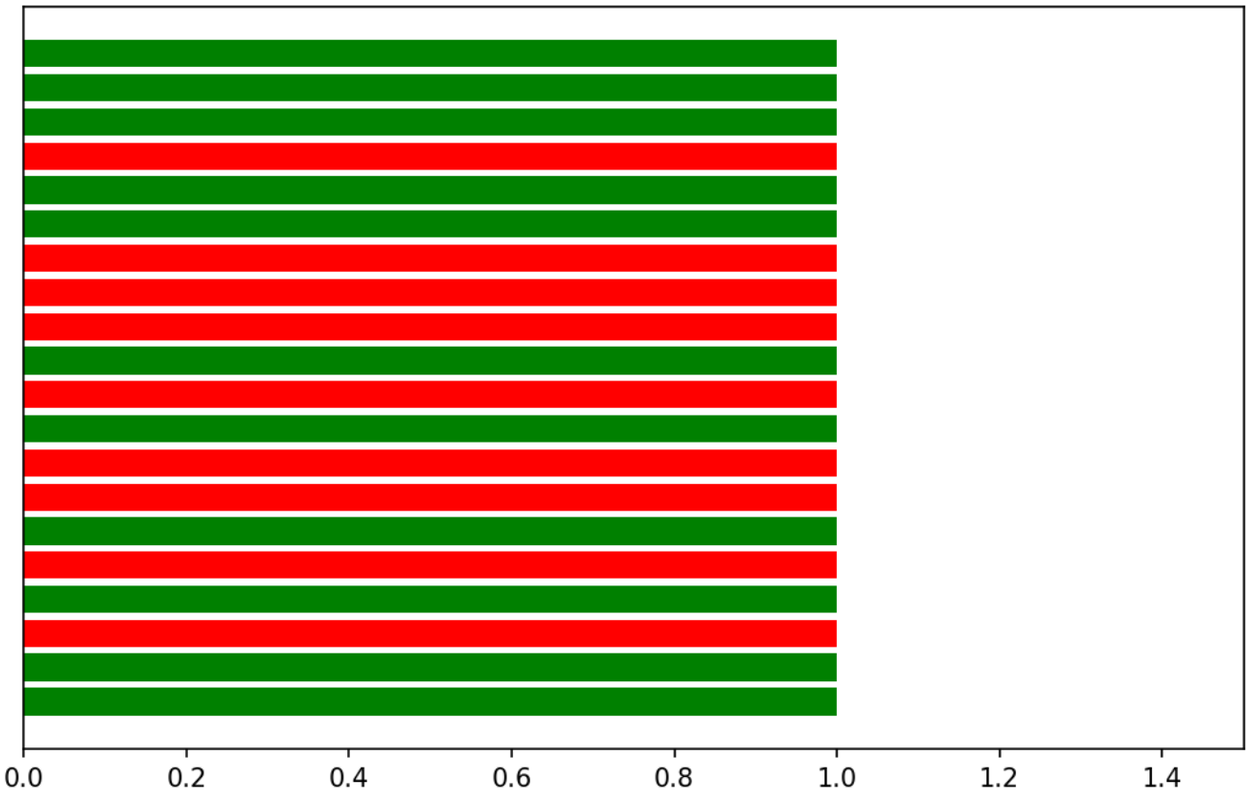

Parameters per Layer
Parameters (Millions)
0
5
10
15
20
25
30
35
Conv1
Conv2
Conv3
Conv4
Conv5
FC1
FC2
FC3

AlexNet Four Key Innovations

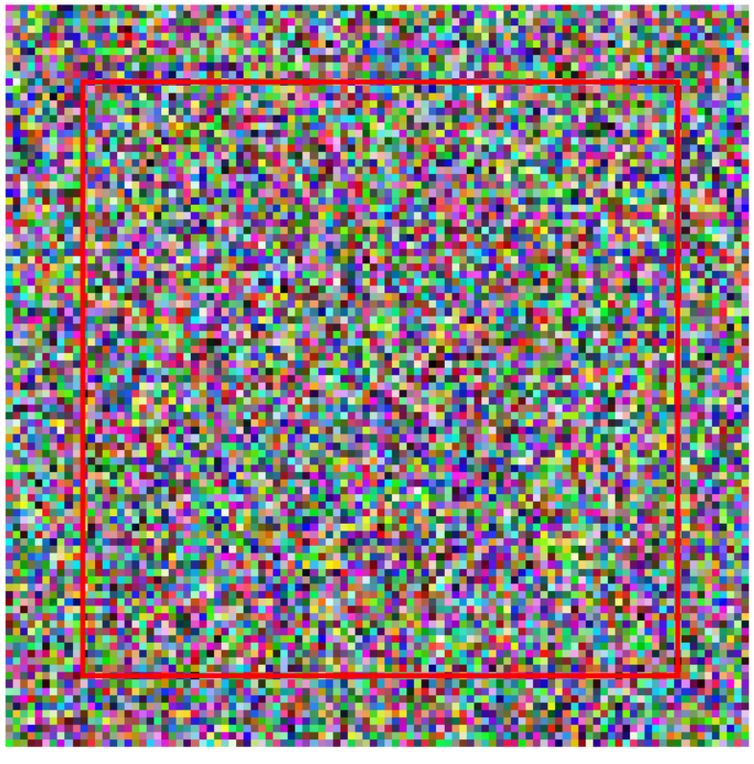
1. ReLU Activation (Faster, Sparse)



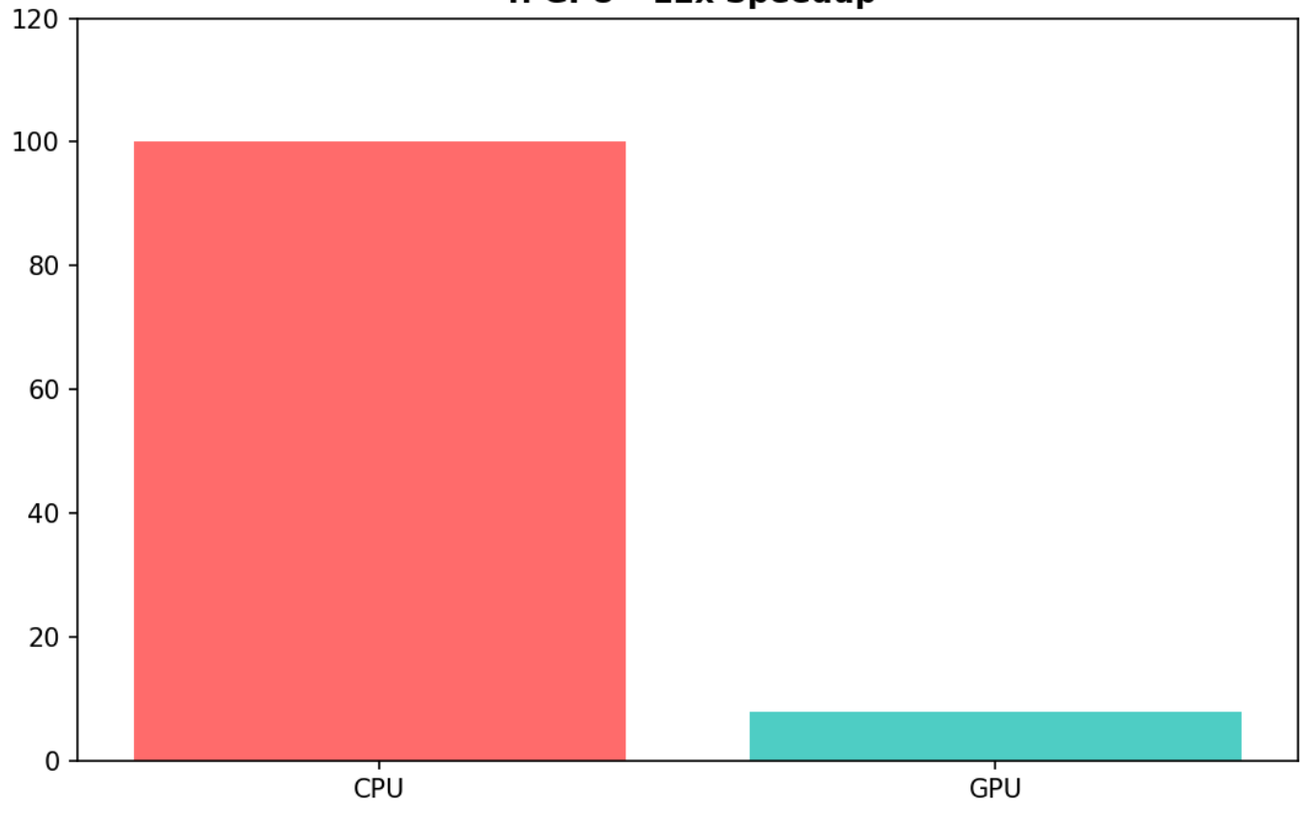
2. Dropout (Random 50% OFF)

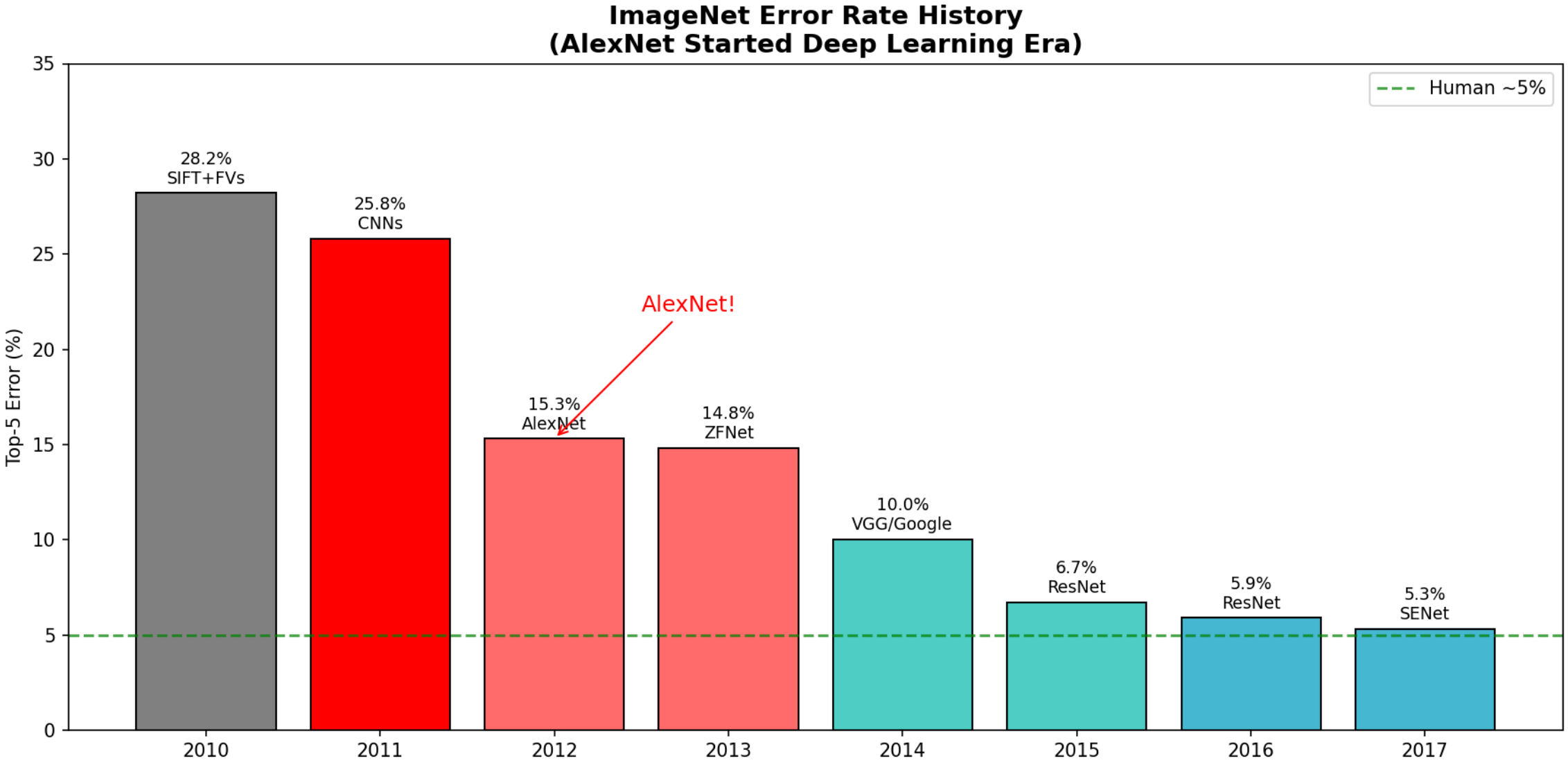


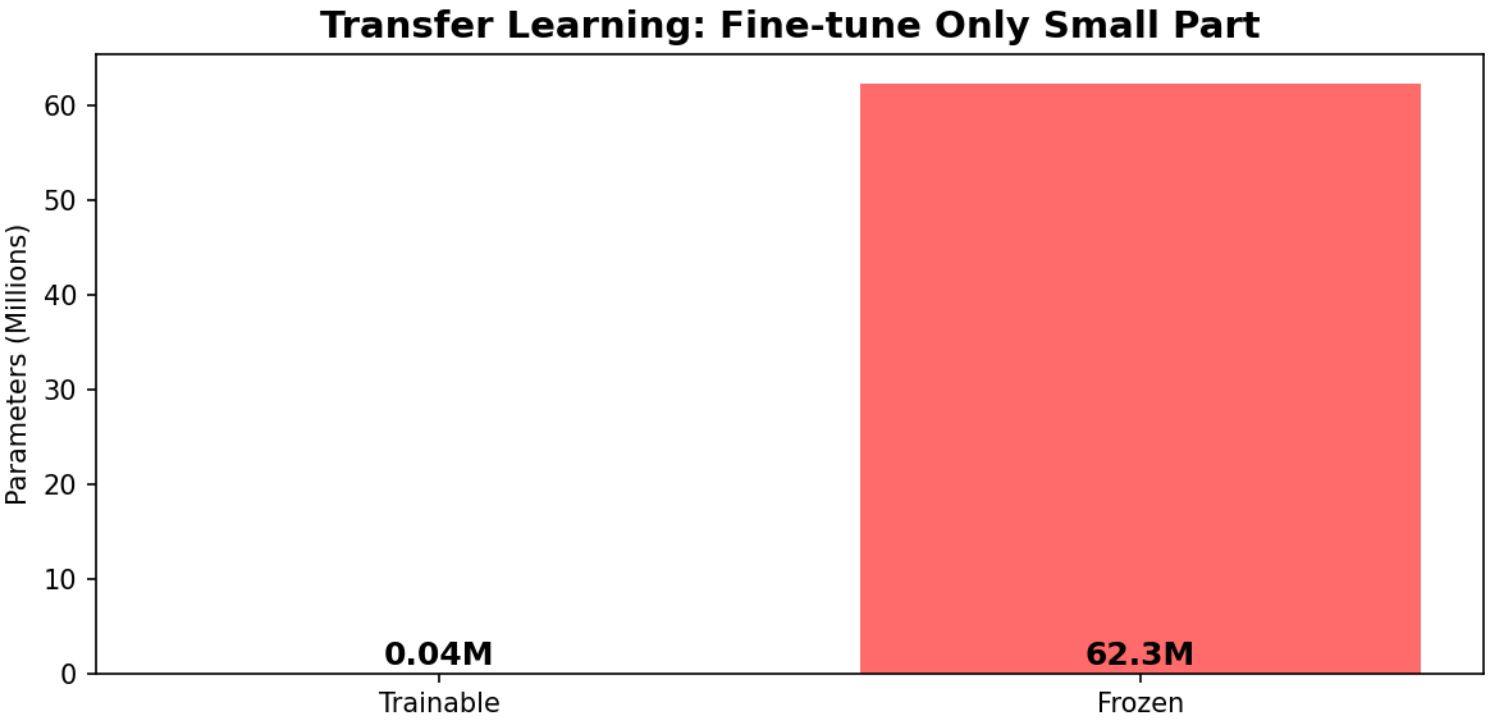
3. Data Augmentation



4. GPU ~12x Speedup




ImageNet Error Rate History
(AlexNet Started Deep Learning Era)
Top-5 Error (%)
35
30
25
20
15
10
5
0
Human ~5%
28.2%
SIFT+FVs
25.8%
CNNs
AlexNet!
15.3%
AlexNet
14.8%
ZFNet
10.0%
VGG/Google
6.7%
ResNet
5.9%
ResNet
5.3%
SENet
2010
2011
2012
2013
2014
2015
2016
2017


Transfer Learning: Fine-tune Only Small Part
Parameters (Millions)
0
10
20
30
40
50
60
0.04M
62.3M
Trainable
Frozen